



UNIVERSITY OF MORATUWA

DEPARTMENT OF ELECTRONIC AND
TELECOMMUNICATION ENGINEERING

EN2160 - Electronic Design Realization

Final Report

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1 Problem Description, Motivation, and Justification

1.1 Problem Statement

In Sri Lanka, there are frequent grid power interruptions disrupting the operation of day to day electronic devices. While some affordable backup power solutions exist, most of the budget friendly inverters that currently exist in the market results in problems such as electrical noise, overheating and long term degradation in sensitive equipments. Hence there is a critical need for a cost effective and high efficient pure sine wave inverter that can deliver clean, grid quality AC power using an existing battery to power even the sensitive electrical and electronics during power outages.

1.2 Motivation & Survey Results

The motivation stems from severe energy instability experienced all over Sri Lanka in recent years. National grid struggling with capacity constraints, repairs for the grid equipments made reliable access to electricity became a daily challenge for households, disrupting remote work, education and essential communication during disasters.

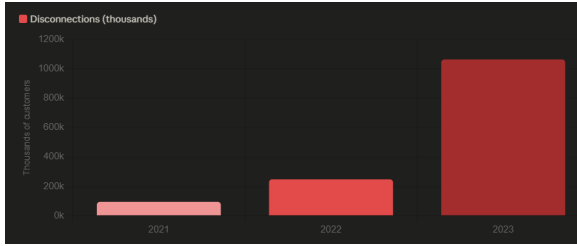


Figure 1: Over 1 Million Sri Lankan households lost power access in 2023.

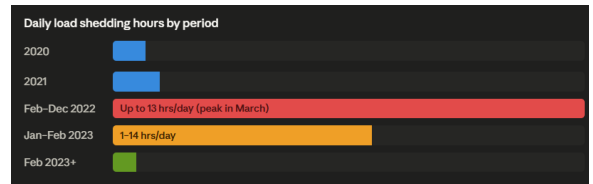


Figure 2: Scheduled power cut hours per day in Sri Lanka.

As illustrated in Figures 1 and 2, the frequent outages highlights a critical vulnerability for the general households. During these blackouts, users need continuous power for essential, low wattage devices such as Wi-Fi routers, monitors which draw under 300W. While the local market has cheap power backup solutions, those mostly consist of modified sine wave topologies which introduce severe harmonic distortion that causes noise and excessive heat dissipation. Hence these statistics not only confirm the sheer volume of affected households but also validates the specific need for a localized, budget friendly and efficient 300W pure sine wave inverter capable of delivering clean, grid-equivalent power to handle day to day lives.

2 Technical Feasibility

We evaluated the technical feasibility of this project in these following domains:

- **Control Architecture and SPWM Generation:**

For the SPWM generation we had two choices: go with a microcontroller and program it to generate the desired SPWM signals with dead-time controls manually or use an ASIC chip that was custom made for this purpose only. We went with

the custom ASIC chip as that would be the most efficient one in terms of power consumption and resources as this was specifically made for this purpose. We chose the architecture of EG8010 which consists of controllable output frequency, dead-time and safety measures for over-voltage and over-current built in.

- **Power Stage and Thermal Management:**

This part is also covered under the architecture used in EG8010 which uses thermistors in its control loop, to turn on a fan after it reaches a certain temperature threshold. We also used high power MOSFETs with custom heat-sinks attached to it so the heat can be dissipated effectively.

- **Component Sourcing and PCB Manufacturing:**

All the components are sourced from **LCSC** with the standard to handle the high power and current requirements. We wound our own ferrite core transformer to accustom to our need. Standard ceramic SMD components were used for non critical components on the PCB. The PCB trace widths are made to withstand the high current and was outsourced from **JLCPCB**. We made 2 PCBs, one is for the controller board which acts as the brain of the inverter and the other one is the power board which has all the critical high current components on it with the transformer being hoisted on the enclosure outside of the PCBs.

- **Testing and Validation Capabilities:**

We tested each board separately, we gave controlled inputs to the control board and observed the SPWM signals generated from the control board, then we connected the controller PCB with the power board with appropriate connections without the transformer and observed a clean pure sine wave at 50 Hz. Then we assembled everything together as one system and connected a standard home appliance checked its performance to ensure our product is working.

3 Product Architecture

3.1 Block Diagram

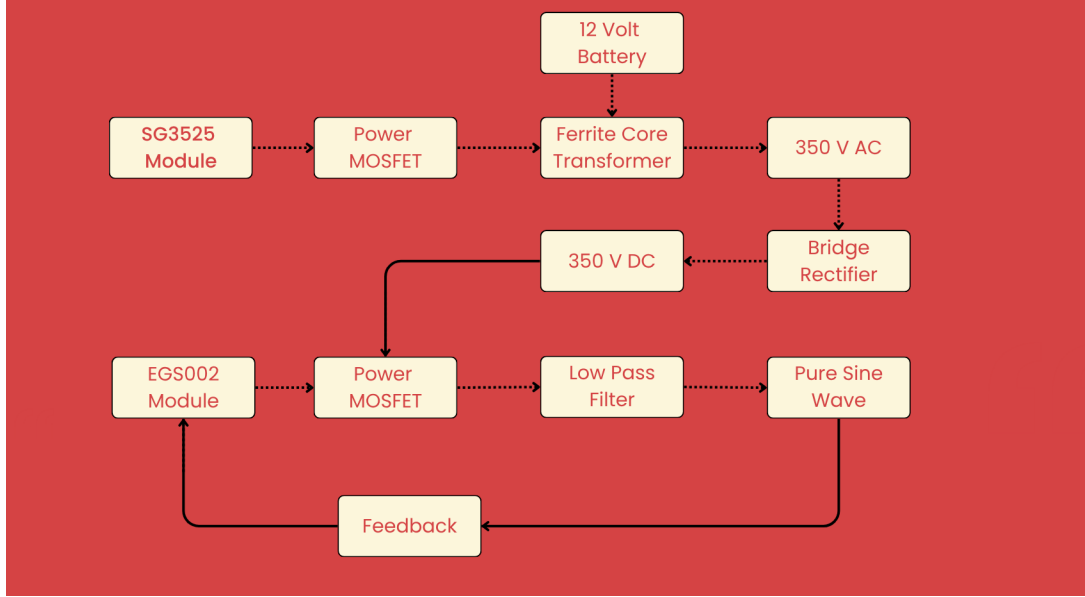


Figure 3: System Block Diagram showing core functionality.

3.2 Component Description & Functionality

Based on the system architecture, the inverter operates using a high-frequency, two-stage power conversion topology. The functionality of each core block is as follows:

- **12 Volt Battery:**

Primary DC power source supplying high-current, low-voltage power to the primary side of the transformer.

- **SG3525 Module & Power MOSFETs (DC-DC Boost Stage):**

SG3525 provides the high frequency PWM drive signals to rapidly switch the primary side Power MOSFETs on and off. This converts the 12V DC into a high frequency square wave.

- **Ferrite Core Transformer:**

A high frequency custom wound ferrite core transformer which steps up the 12V chopped square wave to a high frequency 350V AC.

- **Bridge Rectifier:**

Using fast-recovery diodes configured in a full bridge to convert the high frequency AC back to a high-voltage stable 350V DC bus.

- **EGS002 Module (SPWM Controller):**

Brain of the DC-AC conversion stage built around the EG8010 ASIC architecture, which generates precise SPWM signals with built-in dead-time control to prevent shoot throughs in the final power stage.

- **Power MOSFETs (H-Bridge Inverter Stage):**
Full H-bridge configuration of high voltage MOSFETs, driven by IR2110 driver to rapidly switch 350V DC bus to generate an AC high-voltage SPWM waveform.
- **Low Pass Filter (LC Filter):**
The filter to smooth out the high-frequency switching edges of the SPWM signal to convert into a clean 50Hz 230V AC output for sensitive electronics.
- **Feedback Loop:**
Feedback loop for the voltage consists of a step-down resistor divider using that we continuously monitor the final AC output voltage. This data is feedback to the controller board to adjust the SPWM duty cycle in real-time to maintain a stable 230V 50Hz output even when the load changes or the battery drains.

4 PCB Design

4.1 Schematic and Layout Design

To reduce complexity of the whole system, we divided into two PCBs: a Controller Board and a Power Board. This separation makes sure that the low-voltage control signals are isolated from the high-current, high-voltage switching paths significantly reducing EMI and improving the system stability overall.

Hierarchical Schematic Approach

We followed a hierarchical schematic design approach to reduce clutter when following the schematic for technical reasons. This allows easy debugging and improves readability.

1. Controller Board

The Controller Board houses the low-voltage logic and signal generation components, including the EG8010 chip, alongside the voltage and current feedback conditioning circuits.

- **Layout Considerations:** We prioritized short traces for high frequency signals and utilized a ground plane for better performance.

2. Power Board

This board Contains two main parts, 12V - 400V voltage converter and the Hbridge+Output filter.

4.1.1 12V - 400V conversion

A push pull converter was designed to boost 12V to 400V. It contains the 12V DC input bus, primary-side MOSFETs, DC rectifier and the high voltage H-bridge output stage with the filter.

- The SG3525 was used due to its ability to adjust the duty cycle of its PWM output signal to keep the output voltage (400V) constant under varying load conditions.

This is achieved using a feedback signal obtained by using a voltage divider with a variable resistor.

- The primary winding of the transformer was sized to handle a current of approximately 25A. 24 strands of 0.6mm magnet wire were fashioned into a litz wire.
- **Layout Considerations:** The appropriate trace widths were used for the high current paths in the primary-side. Strict clearance and creepage distances were maintained for the high-voltage sections to prevent high-voltage arcing. Testing points were used in both boards for quick debugging and testing.

4.1.2 Hbridge + Output Filter

This part Converts the 400V DC into 230V sinewave AC.

- Optimized low inductance traces were used for the high voltage DC bus and H-bridge switching nodes to minimize parasitics and to reduce voltage overshoot during MOSFET turn-off.
- The high side and low side MOSFET pairs were placed symmetrically close to the gate driver to keep gate drive loop areas small and minimize switching noise.
- Gate resistors and bootstrap components were placed as close as possible to their respective MOSFET gates and bootstrap diodes to ensure clean, fast switching transitions.
- A solid ground plane was maintained beneath the switching stage to provide a low-impedance return path and reduce EMI coupling into sensitive control traces.
- Strict clearance was enforced around the high-voltage DC bus and switching nodes, to satisfy high-voltage safety margins, also the output LC filter was placed close to the H bridge output to minimize the loop area exposed to high speed switching.

1. Controller Board Schematics

The Controller Board houses the low-voltage logic, signal generation components, and isolated power supplies. To manage the design hierarchically, the circuitry is divided into specific functional blocks as shown below.

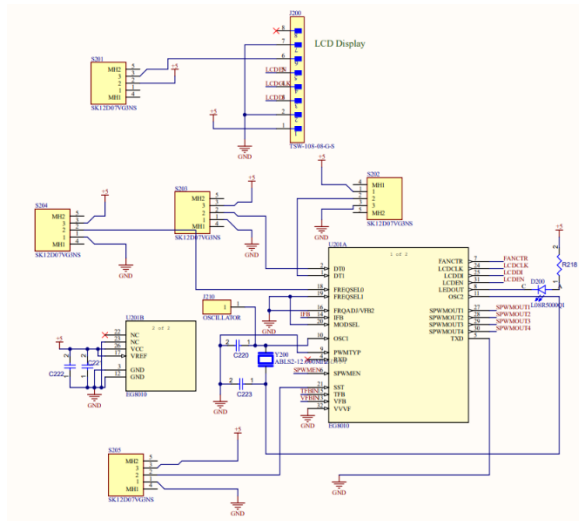


Figure 4: EG8010 SPWM Controller.

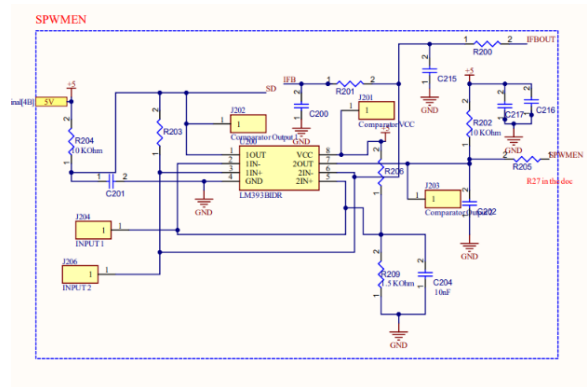


Figure 5: Comparator / Protection Logic.

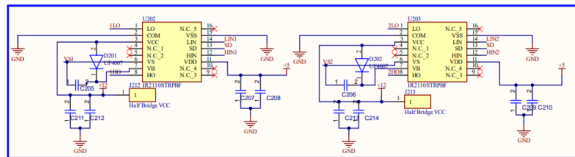


Figure 6: Gate Driver Circuitry.

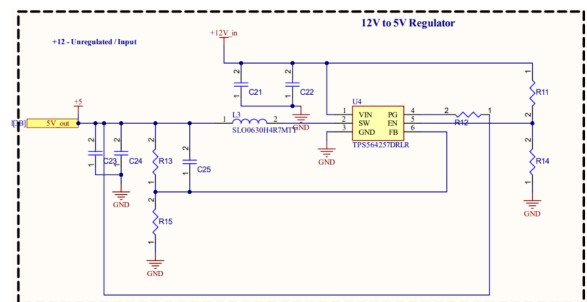


Figure 7: 12V to 5V Logic Power Supply.

Figure 8: Primary control and logic schematics for the Controller Board.

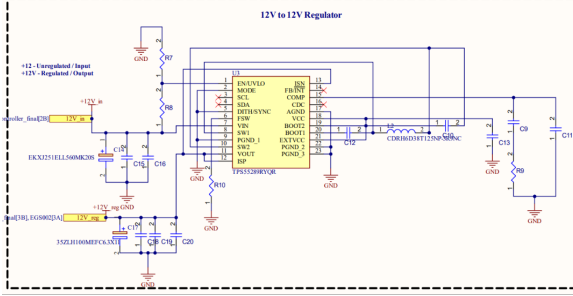


Figure 9: 12V to 12V Isolated Supply.

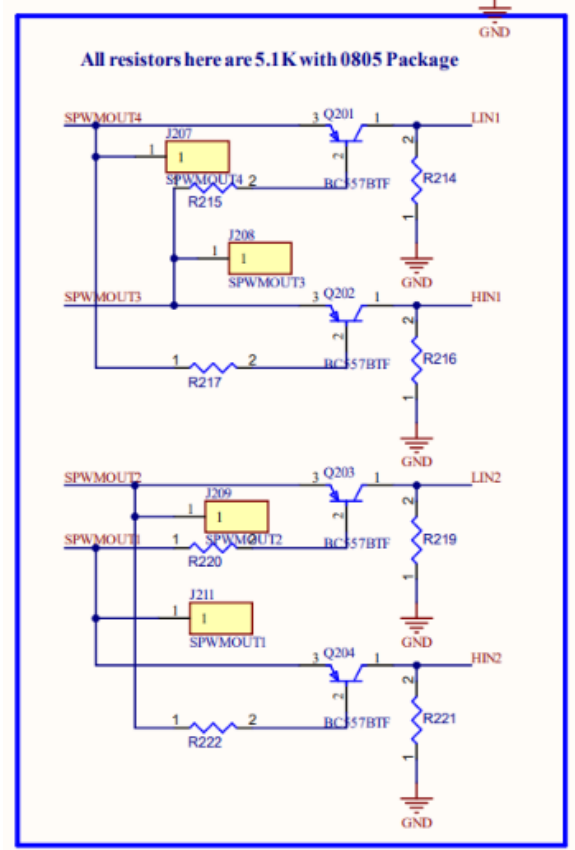


Figure 10: Output Switching Section.

Figure 11: Power regulation and output sections of the Controller Board.

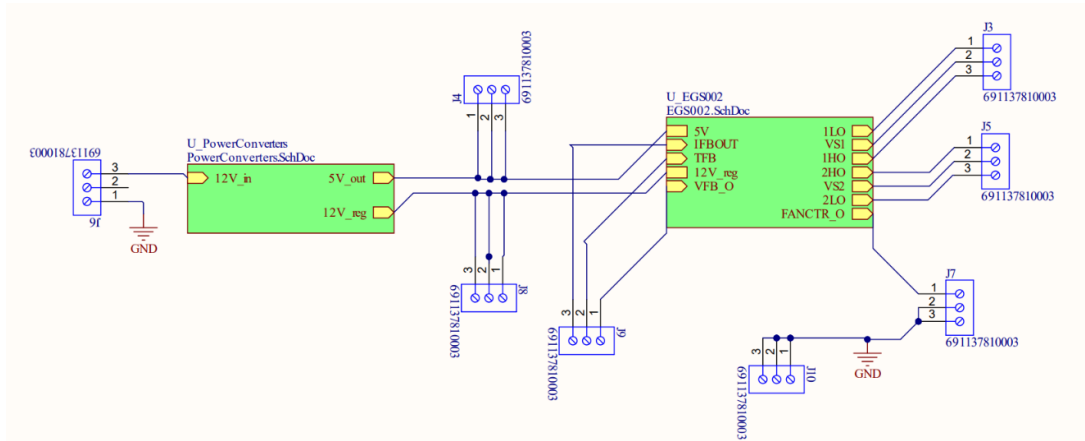
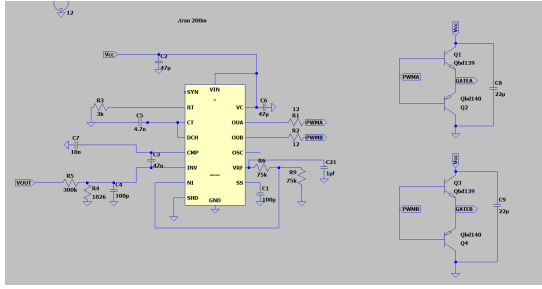


Figure 12: Top-level hierarchical root sheet showing the macro-level interconnections between all subsystems.

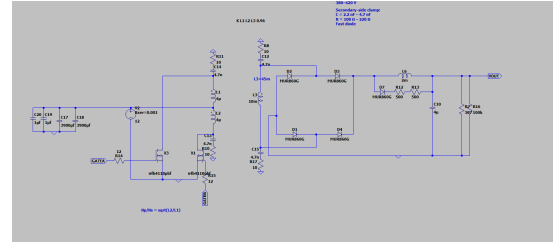
4.2 Push pull converter simulation

A push pull converter stage was used to boost 12V to 400V. A custom transformer with a center tapped primary was designed by ourselves. The winding ratio calculated was 1:40. LTspice was used to simulate and verify the values required for the passive

components(Resistors, capacitors) of the SG3525. Transformer winding ratios, current draw, RC snubber circuits were tested and implemented using the simulations.

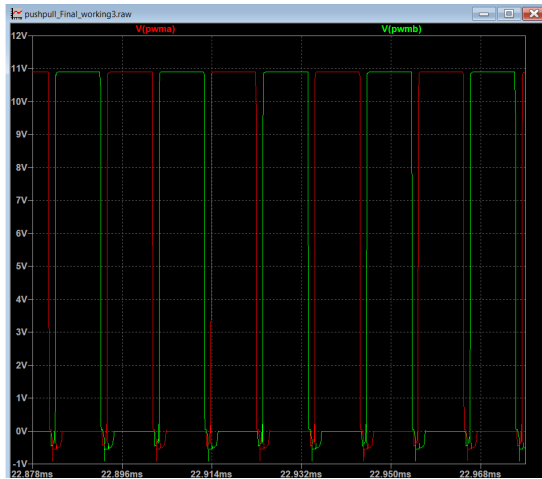


(a) SG3525

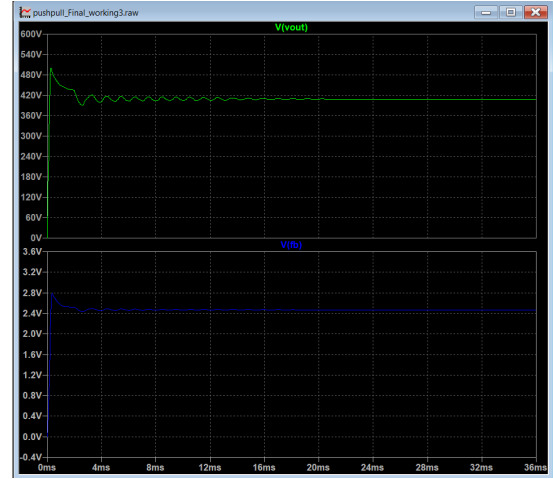


(b) Transformer

Figure 13: 12V to 400V



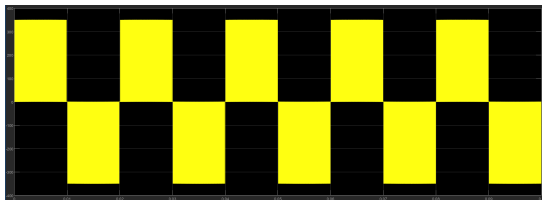
(a) Mosfet gate drive signals



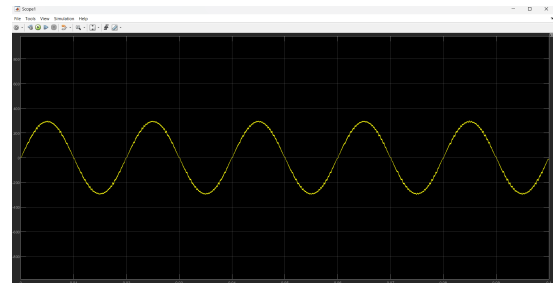
(b) Output Voltage and Feedback voltage

Figure 14: 12V to 400V

4.3 Hbridge & Output simulations



(a) Unipolar PWM output from the Hbridge



(b) Final Output Voltage after Filter

Figure 15: 12V to 400V

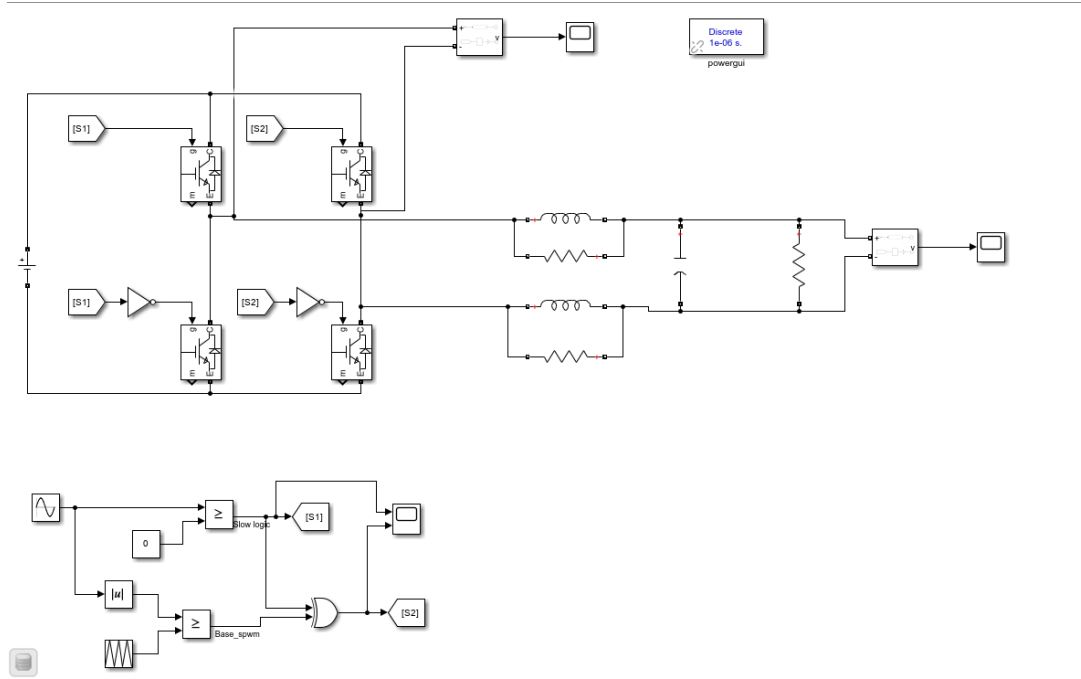


Figure 16: Simulink model

4.4 Power Board Schematics

The Power Board handles the bulk power transfer and voltage conversion. Its hierarchical design is divided into six functional modules to ensure clear signal routing and effective thermal management.

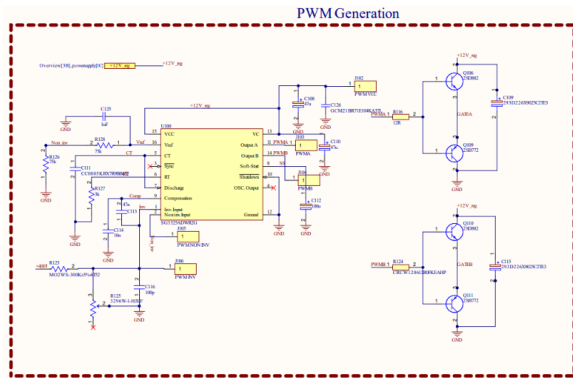


Figure 17: PWM generation and totem pole driver.

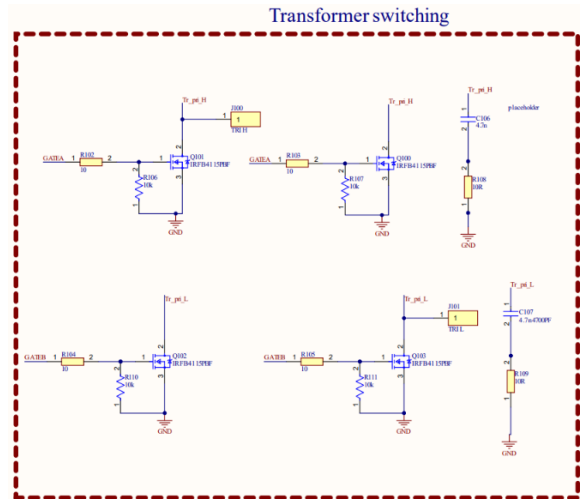


Figure 18: Transformer Switching.

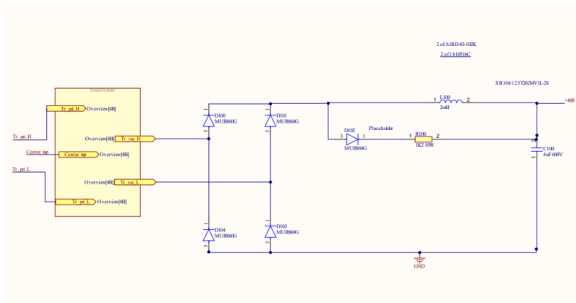


Figure 19: Full wave Rectifier Section.

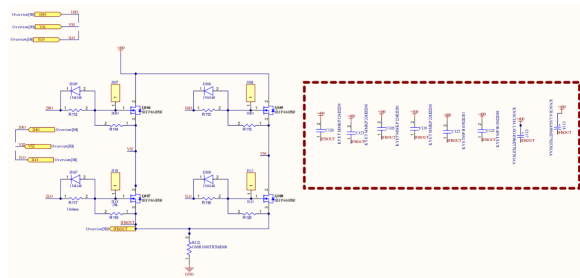


Figure 20: H-Bridge Inverter Stage.

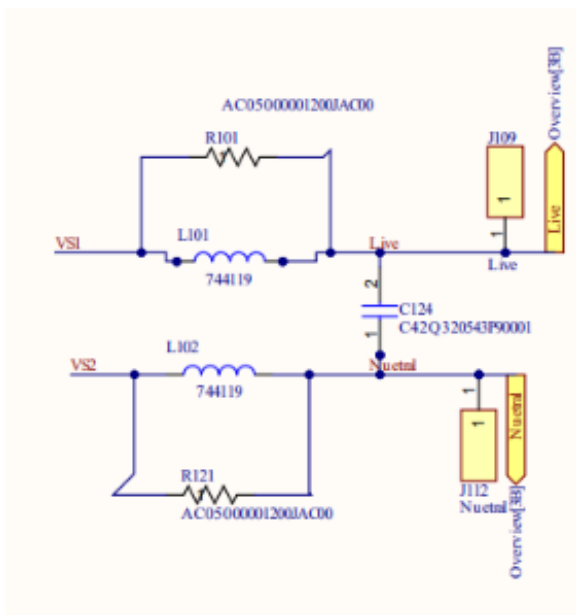


Figure 21: LC Output Filter.

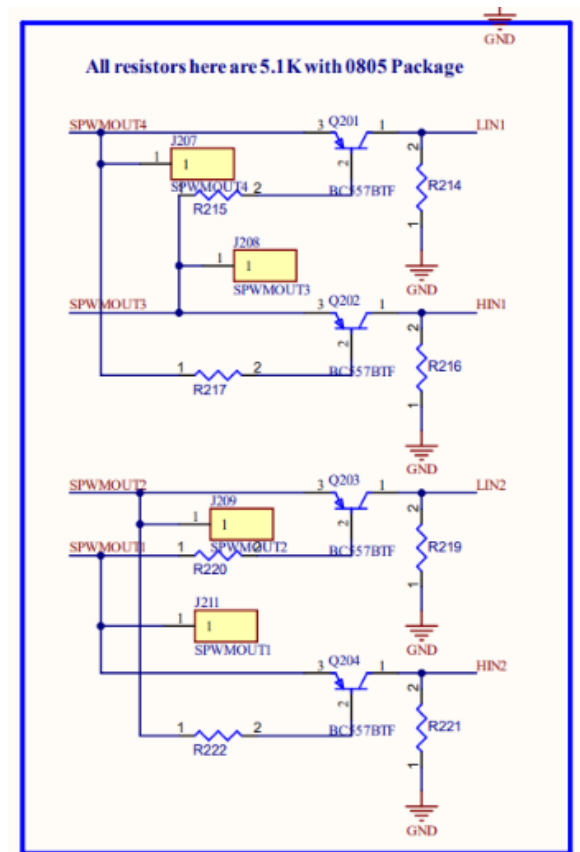


Figure 22: Output Switching Section.

Figure 23: Hierarchical schematic blocks comprising the Power Board.

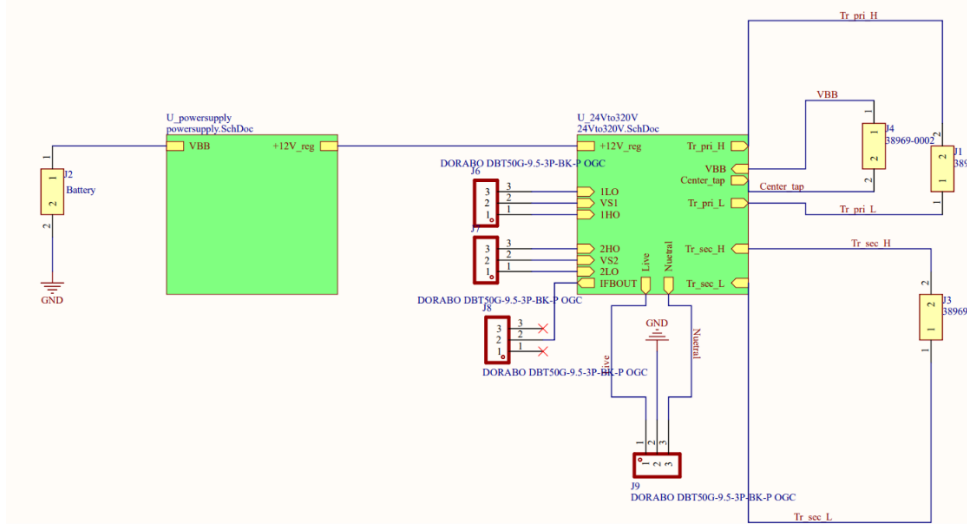


Figure 24: Power Board Hierarchical Root Sheet.

4.5 H-bridge & filter Simulations

Verification of the Bridge implementation and the design and the Component selection for Output Filter was done using MATLAB Simulink.

4.6 PCB Stack Justification

Both boards were designed using a standard 2-layer FR4 stack-up with a 1.6mm thickness and 1oz copper weight (JLCPCB).

- **Why 2-Layer:** A 4-layer board was deemed unnecessary and an overkill for this prototype. To be cost effective we went with 2 layer 2 PCBs.
- **Copper Thickness Compensation:** While 2oz copper is the ideal one for the high-current boards, we went with the 1oz copper due to budget constraints. But still we utilized appropriate trace widths to increase the current carrying capacity.

5 Product Enclosure

5.1 Enclosure Design

We designed the enclosure prioritizing thermal management, safety and easy debuggability. Initial design concepts were hand drawn to visualize the placement of the PCBs and other equipments.

initial sketch (match box)

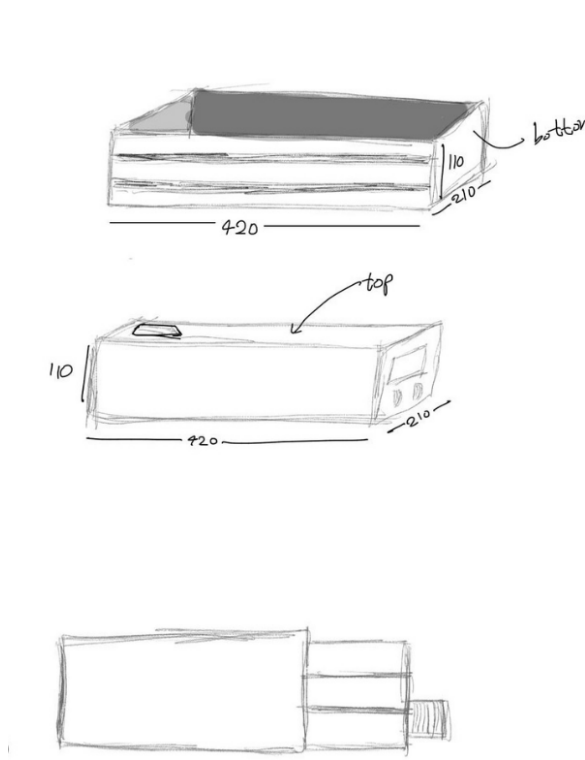


Figure 25: Initial conceptual sketch of internal layout.

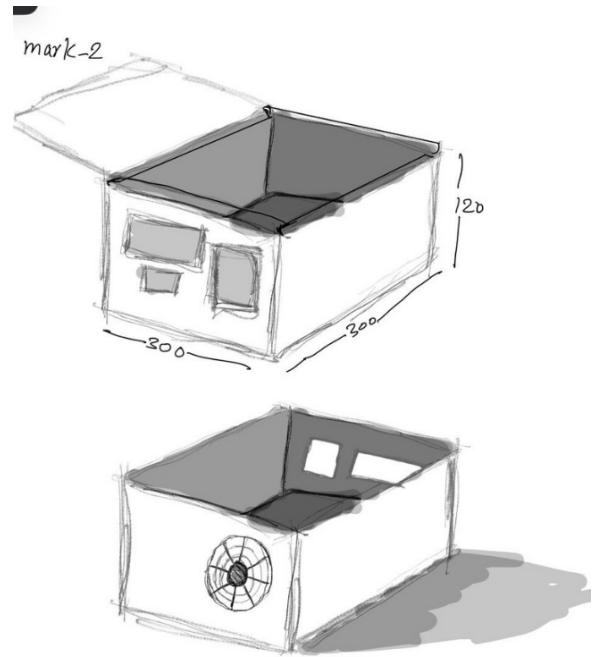


Figure 26: External port and cooling ventilation design.

Figure 27: Hand-drawn sketches detailing the enclosure architecture.

5.2 Finalized Design

The finalized enclosure is designed to house the PCBs, Fan, Transformer, Heat sink and other connecting equipments. They are positioned in such a way that the thermal management is efficient. The enclosure was initially to be made up of mild steel but the metal couldn't be bent the way the enclosure was designed. So the next choice we went with was zinc coated steel which was perfect to the budget and has good qualities.

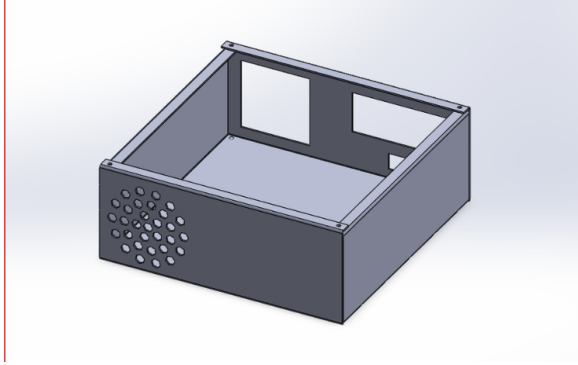


Figure 28: Rear view showing terminal and I/O layout.

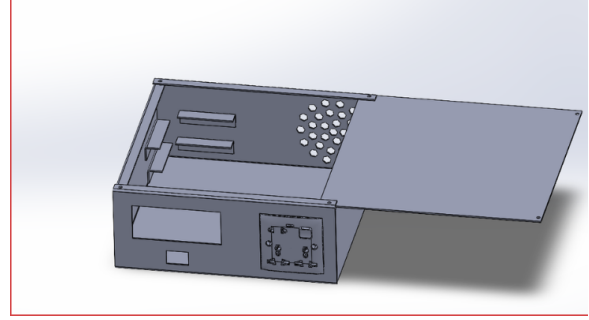


Figure 29: Front view of the finalized enclosure.

Figure 30: Finalized design of the inverter enclosure.

6 Final Product Details

6.1 System Overview

The final product integrates both Controller and Power PCBs into a metal enclosure. Here we outline all the important details of the 300W Pure Sine Wave Inverter.

6.2 Physical Assembly and Integration

All the components are carefully integrated into the enclosure ensuring electrical safety and optimal thermal performance.

- **Board Interconnection:** All the boards are connected with appropriate wires matching their specifications (current rating). Thick wires were used to wind the transformer to match our need.
- **Component Placement:** Inside the enclosure each PCB is mounted on a 3D printed hoist to maximize thermal dissipation and they are separated via space to ensure isolation between high-voltage sections and low-voltage control sections.
- **Mounting and Securing:** Square Socket is used to connect home appliances to the inverter. 30A mounting post is used to connect a low-voltage, high-current battery to the input of the inverter. A small switch is used to turn the inverter on. A small mount is fitted to the front side of the inverter to hold the display which will show real-time values.

6.3 Bill of Materials and Manufacturing Summary

Details of key components and manufacturing processes used for the final product:

- **Active Power Devices:** High-current Power MOSFETs, and fast switching diodes were sourced from LCSC which can be bought for a cheaper price when buying in bulk. The center tapped ferrite core transformer is built in house by us to match the needs for this inverter.

- **Control Logic Components:** EG8010 SPWM generator and SG3525 PWM controller is outsourced from Sri Lanka. SMD versions of the driver modules is outsourced from LCSC as well.
- **Fabrication Methods:** Two-layer FR4 PCBs x2 is outsourced from LCSC with optimized 1oz copper traces.

6.4 System Testing and Validation

Testing was done in parts, the controller board and the power board was tested separately first then both boards were connected and tested without the inclusion of the transformer, then both boards were tested with the inclusion of the transformer. And as the final test our product was tested with a home appliance to ensure the performance of the inverter.

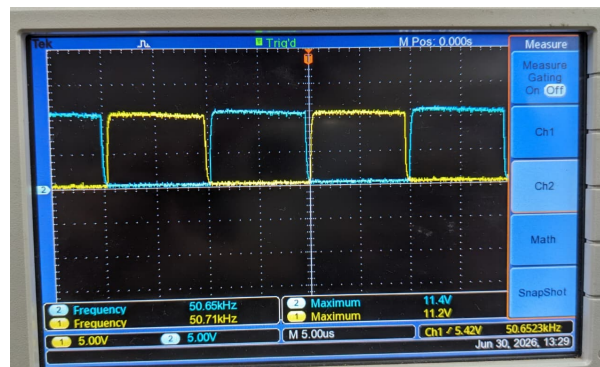


Figure 31: Mosfet gate drive signals from the SG3525



Figure 32: Top View of the Final Product

7 Manufacturing Considerations

7.1 Overview

Transitioning from a prototype into a mass production is a whole different ball game. Since our product is a power product it requires rigorous manufacturing and assembling strategies. Special care must be given to high-voltage isolation, heavy-current copper traces and structural integrity of the enclosure.

7.2 Custom Transformer Fabrication (In-House Winding)

For the prototype we wound our own transformer matching our needs. For mass production this responsibility can be outsourced to a manufacturer to get high quantity, high quality transformers to be used for manufacturing.

- **Core Selection and Preparation:** Proper stacking and gluing of the ferrite/laminated iron core to eliminate air gaps where possible and minimize audible buzzing or core vibration under load.

- **Wire Gauge (Skin Effect and Current Density):** Utilizing multi-strand magnet wire (litz wire technique or bundled enamelled copper wire) for high-frequency windings to mitigate skin effect losses, alongside appropriately rated heavy gauge wire for primary high-current loops.
- **Layer Insulation:** Application of high-temperature electrical grade insulating tape (e.g., Kapton or Nomex) between primary and secondary layers to guarantee dielectric breakdown strength and prevent high-voltage flashover.
- **Tension and Symmetry:** Ensuring uniform winding tension to maintain tight coupling coefficients between primary and secondary coils, reducing leakage inductance which otherwise causes damaging voltage spikes on the switching MOSFETs.

7.3 Enclosure Fabrication (Zinc-Coated Steel)

The enclosure was fabricated from zinc-coated steel, which can be mass produced by outsourcing to a proper manufacturing company which can provide distinct mechanical and electrical advantages.

- **Structural Rigidity and Durability:** Zinc coated steel gives good mechanical strength and protection for heavy internal components and has a good overall structural integrity.
- **Corrosion Resistance:** The sacrificial zinc layer prevents oxidation and rust formation, ensuring long-term reliability even in humid environments or industrial operating conditions.
- **Electromagnetic Interference (EMI) Shielding:** The steel acts as a natural faraday cage which significantly reduces EMI from high-frequency switching stages and protecting sensitive equipments.
- **Grounding and Safety Integration:** The metallic nature ensures proper fault-current handling and improves user safety during insulation failures.

7.4 PCB Fabrication and Assembly

- **Trace Width and Copper Weight:** Using 1oz copper layers with expanded polygon pours to safely conduct primary currents without excessive resistive heating.
- **Component Placement and Clearance:** Maintaining strict physical separation between the low-voltage control circuitry (EG8010 and ICs) and the high-voltage DC/AC switching stages to minimize electromagnetic interference (EMI) coupling. Creepage and clearance distances were intentionally maximized around the 350V DC bus lines.
- **Thermal Interface Management:** Application of thermal paste to etch the heatsink with the power mosfets to guarantee low thermal resistance.

8 Costing and Pricing

8.1 Production Cost Analysis

8.1.1 Prototype Costing

- **Marginal Cost:** Rs. 14,000.00
- **Absorption Cost:** Rs. 16,000.00 (*including transport and other non-manufacturing costs*)

8.1.2 Costing at Volume Production

Based on our analysis of the market and supply chains, unit costs can be reduced through economies of scale as follows:

Table 1: Projected Cost Reduction at Volume Production

Component	Cost Reduction
PCB	20%
Enclosure	30%
Other Components	10%

With these reductions applied at volume, the per-unit product cost can be brought down to **Rs. 10,000.00**.

Finally, including marketing, manufacturing, and transport costs, the total per-unit cost at volume production would be **Rs. 13,000.00**.

Table 2: Competitor Landscape

	Tier 1 — Pure Sine Wave	Tier 2 — Hybrid Solar	Tier 3 — Generators
Input	12V DC battery	Solar + Grid	Petrol / Diesel
Brands	Unitec, SAKO, Growatt, Sorotec	Goodwe, Huawei, Victron, Growatt Hybrid	Yamaha EF, Honda EU, Local brands
Advantages	Clean output, portable, locally available	Solar integration, smart monitoring, long warranty	High power, no battery needed, widely available
Disadvantages	Import markup, limited local service, 110V/60Hz mismatch	Overkill (3–10kW), very expensive, fixed install only	Fuel cost, loud, heavy, exhaust emissions
Price Range	LKR 25,000–40,000	LKR 200,000+	LKR 40,000–400,000

Our product has direct competition with Brands like Unitec, SAKO, GOrowatt, Sorotec. But since these brands are imported they have a tax, also less reliable services and voltage and frequency mismatches. Other inverters available in the market are

overkill for our target customers with regard to power requirement, features and very expensive.

Hence We are pricing our product at Rs25,000.00.

Profit margin of RS.12,000.00

9 Marketing, Sales, and Beyond

Our product positions itself as a low power Pure sine wave inverter, We target customers who need electricity for basic needs when the grid is down and customers require electricity at off grid places. 300W power level makes the product more affordable than the high power inverters and since this is a pure sine wave AC output it's save to connect sensitive equipments such as laptops.

- House Owners
- Vehicle Owners
- Small business
- Hikers and Campers

Acquiring Customers With the frequent power cuts in the country, demand for inverters is already high. Hence we do a marketing campaign doing advertisements highlighting Low power requirements, affordability and Sri lankan made design and manufacturing. In the early stages of the product launch we offer discounts and renting services to get customers interested. Also getting a local brand ambassadors such as a youtuber related to electric appliances. We are also planning to approach local companies like hayleys electronics once we obtain considerable market share.

Warrenty period of 1 year, and two free services for the inverter will be provided.

10 Project Budget and Bill of Quantities (BOQ)

Table 3: Bill of Quantities (BOQ) for Prototype

Item	Description / Part Number	Qty	Unit Cost (Rs.)	Total (Rs.)
1	EG8010 SPWM controller	1	1,200	1,200
2	SG3525 PWM controller	1	250	250
3	IR2110 high/low-side driver	2	300	600
4	Power MOSFETs (H-bridge + push-pull)	6	350	2,100
5	Ferrite core transformer (custom wound)	1	1,200	1,200
6	LM2678SX-5.0 buck regulator	1	350	350
7	TPS55289 buck-boost converter	1	500	500
8	Bridge rectifier + filter capacitors	1 set	850	850
9	LCD display module	1	650	650
10	Connectors, resistors, misc. passives	1 set	6,000	6,000
Total Prototype Cost			Rs. 13,700	

11 Future Improvements

11.1 Overview

While our product satisfies the purposes of a minimum viable product, it can be further enhanced by using these:

11.2 Potential Enhancements

- **Advanced Control and Monitoring:** Instead of using an ASIC we can use a micro-controller with Bluetooth / Wi-Fi capabilities to make our product smart to transition into an IoT device.
- **Magnetic Component Refinement:** By manufacturing the transformer in an industrial setting, we can optimize the loss caused by the transformer.
- **Thermal and Weight Reduction:** The PCB layout and the enclosure design can be further optimized to reduce the overall size and weight of the device to make it more compact and carry-able.

12 Task Allocation

Team Member	Assigned Tasks & Responsibilities
ATHUKORALA U R	Schematics, Testing and Assembly
DE MEL D J	Power Stage(12V-400V) design, simulation and schematics, PCB design, Assembly and Testing
FERNANDO W H D	Enclosure design, Assembly and Testing
RAHUL B	Controller Design & Schematics, PCB design and routing, Component Selection, Assembly and Testing
RATHEESHAN A R	Controller Design & Schematics and H-bridge & Filter simulation, PCB design and routing, Assembly and Testing